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Asset-based valuation

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What we still have to cover …

$$V = V_{OP} + V_{Other}$$

Typical „other” assets

Surplus cash

„Cash not needed for the operation”

Securities

Stocks, bonds, options. Use the market price.

Investments

Strategic: synergies, consolidation

Unused assets

Assets not used by own operation

Over- and underfinanced pension plans

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What makes an asset? – Classic Accounting

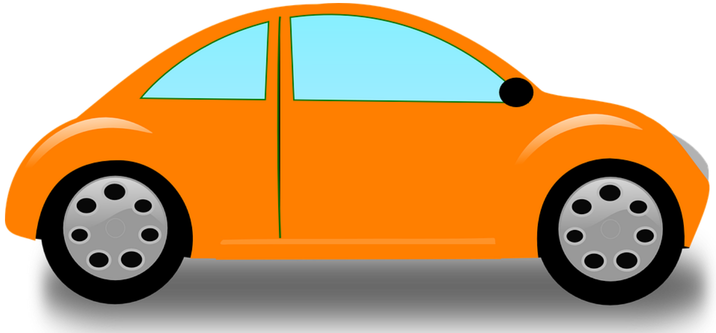
Owned by the firm

Expected incomes exceed expected costs

Nearly no uncertainty

SFAS (until 2009) See Munter, P., & Ratcliffe, T. A. (1980). On the Attributes of an Asset. *Abacus*, *16*(2), 116–123. https://doi.org/10.1111/j.1467-6281.1980.tb00091.x

What makes an asset? – Modern Accounting



Controlled by the firm

Will exchange for cash in the future

Manageable uncertainty

IFRS: "An asset is a present economic resource controlled by the entity as a result of past events. An economic resource is a **right** that has the potential to produce economic benefits."
US GAAP: "An asset is a present **right** of an entity to an economic benefit."

What makes an asset? - Finance

Assets generating cash flows

DCF

Assets not generating cash flow

Replacement value, comparable assets

Option-like assets

Real options

Quasi assets

Asset manageable but not ownable

Asset valuation methods

Type of value	Time orientation of data used		
	Past	**Present**	**Future**
Entry	Purchase cost	Substitution value (repurchase, recreation) Replacement value	–
Exit	–	Net realisable value, Liquidation value	DCF, Capitalisation, Real options

Based on Wilson (1986)

What is our warehouse worth?

We have bought it five years ago for 5 M HUF.

It was renovated last year for 2 M HUF.

Current book value is 5.6 M HUF.

A similar one can be bought for 5.9 M HUF in the neighbourhood.

Last week, 6 M HUF has been offered for it.

We could build a similar one over 18 month for 7.1 M HUF.

The present value of future rent income is 7.4 M HUF.

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Valuation issues

Valuation issues 1 – Current Assets

- Non-operating cash (missing from the DCF value)
- Securities (revaluation)
- Receivables (payment risk)
- Other lending (will we get that back ever?)
- Inventory (FIFO, LIFO, guaranteed purchase price)

Value of cash and securities

Generally, these investments have an NPV of zero, but those could be worth less.

- Double taxation
- Investments below required rate (effective market guarantee zero NPV only at the time of the purchase)
- Investor anxiety about poor M&As (e.g. take-over protection)

Valuation issues 2 – Invested assets

Intangibles: R&D, brand name, franchise, achieves, lists, transferable contracts (purchased vs created, going concern vs liquidation)

Financial assets: shares, bonds, futures, forwards, options, warrants, convertible bonds, convertible shares, swaps, swaptions

Stakes in other firms

Valuation could be challenging (no data):
- market price
- relative pricing
- simplified DCF (target firm cost of capital needed)

<u>May be valued below their market value …</u>

Information gaps (structure not transparent)

Market scepticism

Is the mother company able to manage the cross-owned portfolio optimally? The holding discount may even amount to 5-10%.)

Plant, Property & Equipment

2. Invested assets (cont.)
   c) Machinery
      Real obsolescence (vs D&A):
      1, economic (market situation),
      2, operational (wear & tear),
      3, functional (technology)
   d) Real estate (long useful lifetime, obsolescence, market need, inflation vs D&A)

Consider tax effect if asset sale is assumed.

Valuation issues 4 - Liabilities

- Payables (for each BU)
- Lease (maintenance cost as expense)
- Non-consolidated affiliates
- Liabilities to employees
- Shareholder loans

Valuation issues 4 – Off Balance Sheet

Pension funds (asset or liability)
Postponed tax payments (equity or liability?)
Guarantee, warranty, suretyship
Environmental obligations

Additional challenges

- Litigations
- Compliance
- Special challenges
  - Preferred equity, convertible equity, various types of shares

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The aim of asset valuation is to estimate the amount of the invested capital.

Valuation methods used to estimate capital employed by business units

Net book value	87%
Replacement value	4%
Gross book value	3%
Market value	1%
Other	5%

Source: Arnold-Davies (2000, p. 159.)

Proportion of book value in % of the market value for US listed firms

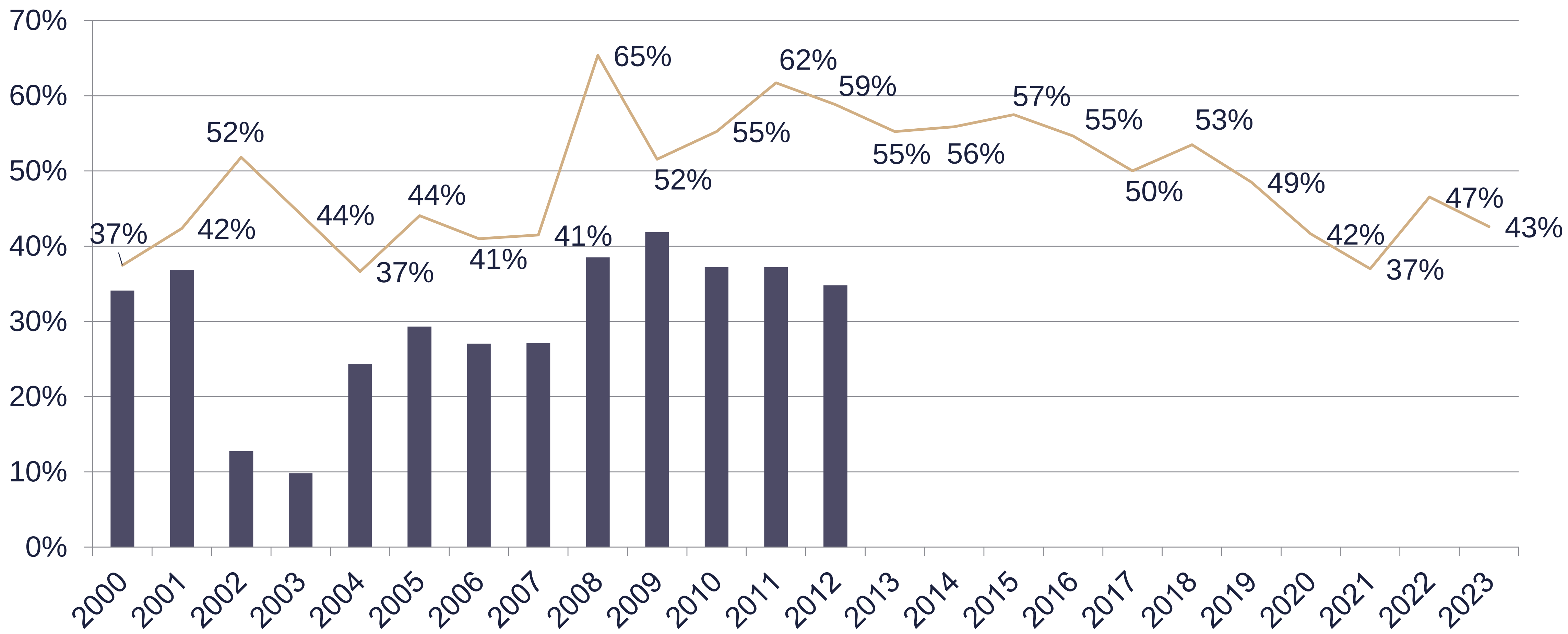
1978	95%
1988	**28%**
2002	**< 20%**

Source: Personnel Today, 2002

Hungary

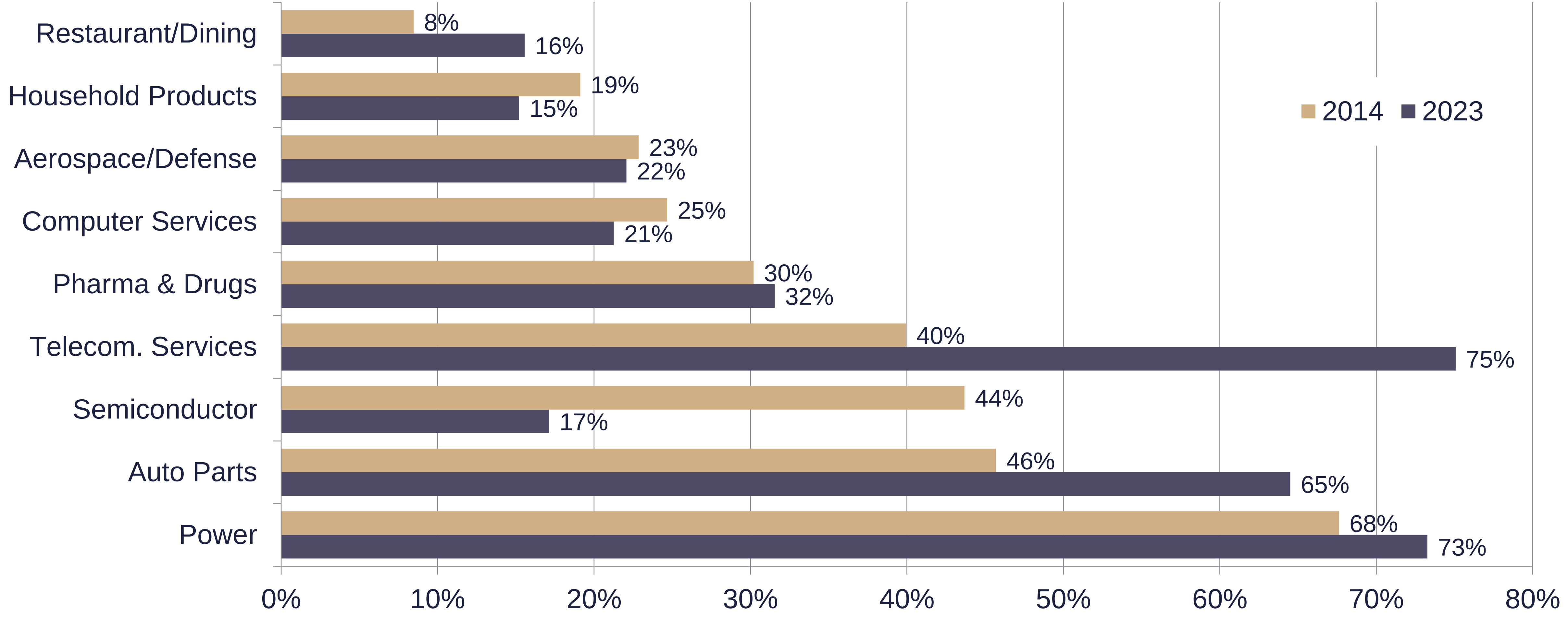
	2004	2009	2013	2019
Net book value	59.3%	53.3%	45.8%	60.7%
Replacement value	5.5%	3.2%	12.0%	3.8%
Gross book value	12.3%	17.8%	7.3%	15.6%
Market value	22.0%	23.6%	33.3%	19.0%
Other	1.0%	2.2%	1.6%	0.9%

The ratio of the book value and the market value of the invested capital in the US*



*December
Source: based on Damodaran (2024)

IC/EV in US*



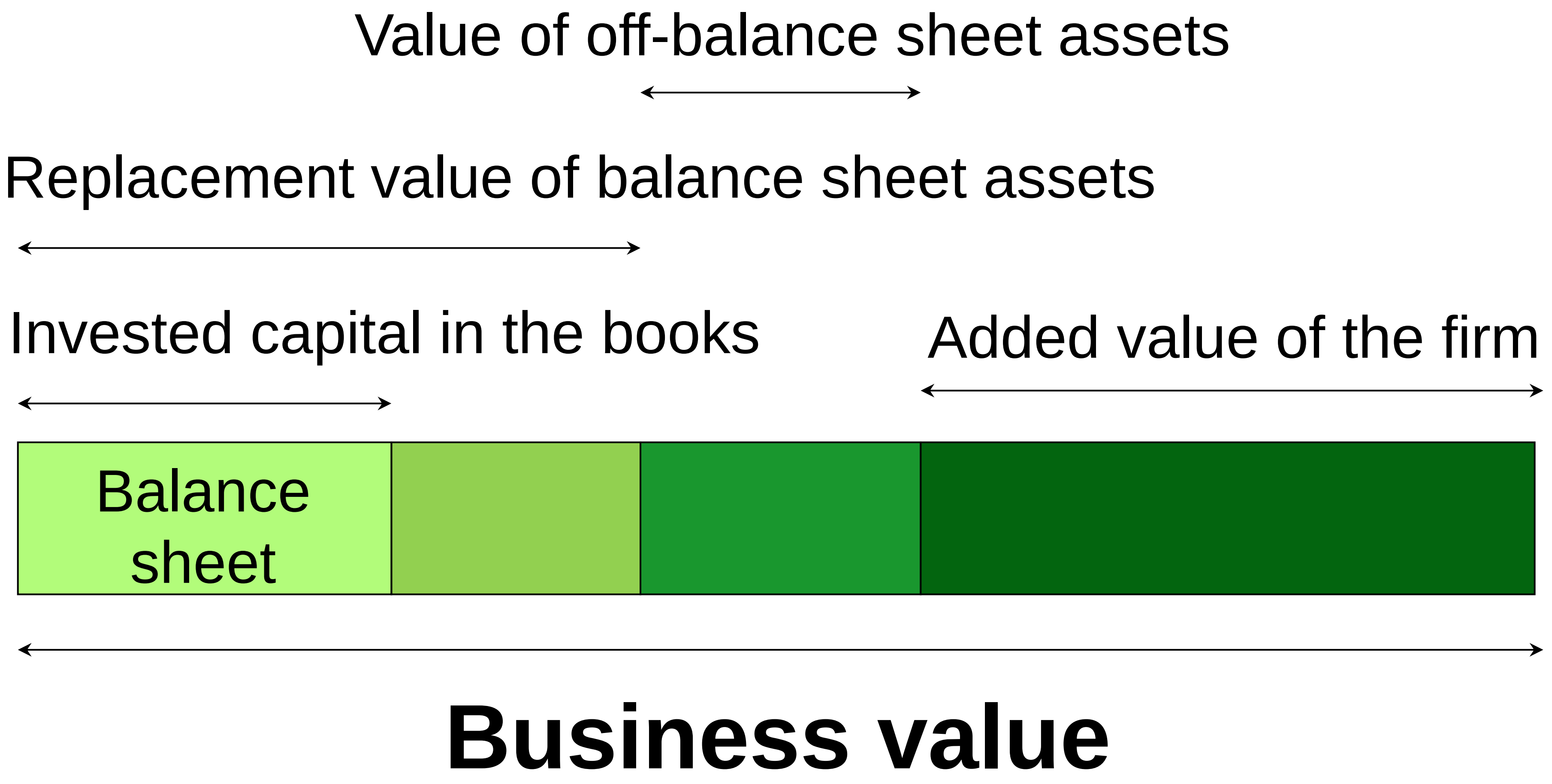
*December

Source: based on Damodaran (2024)

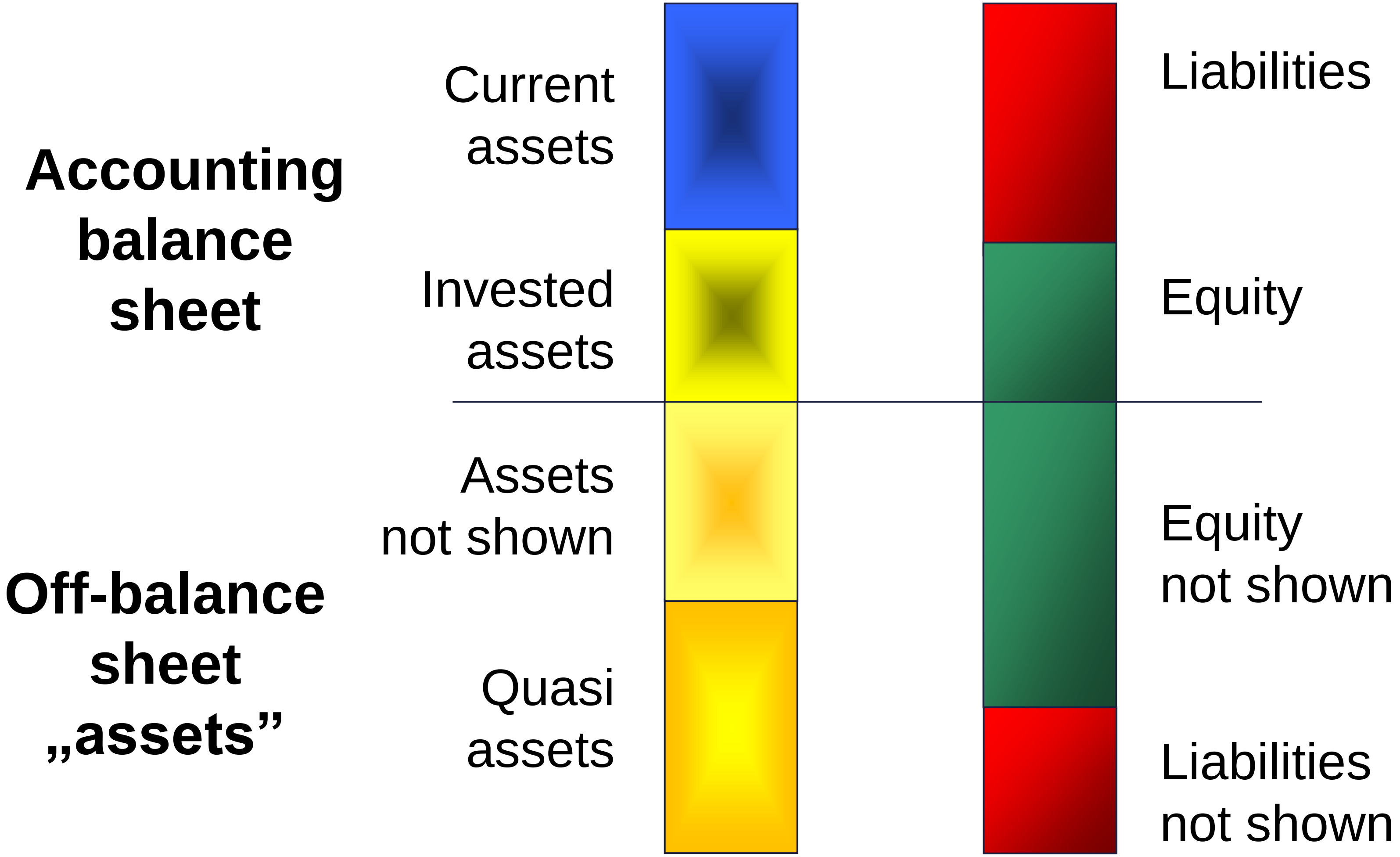
Value perceptions

Accounting value	Business value
Past oriented	Present and future focused
Based on documents	Predictions, estimations
Rule of prudence	Expected value
Aims to inform outside stakeholders	Made for owners and managers

Types of off-balance sheet items



The extended balance sheet



Grouping off-balance sheet items based on type and connectedness

	Ownable	Manageable only
Organisation linked	Brands, transferable contracts	Supply contracts, documented knowledge
Employee linked	– (Playing rights)	Work connections, non-recorded knowledge

Stakeholder	Advantage of good reputation
Investors	Lower price volatility, higher quotes
Employees, management	Lower recruitment cost, higher loyalty (less fluctuation), lower wages
Creditors	Lower rates, more attractive credit terms
Suppliers	Reduced prices, longer payment terms
Retail chain	Lower fees, preferential treatment, wider spatial coverage, more flexible connection
Customers	Premium price, more frequent purchases, higher volume
Competitors	Entry barrier, comparative advantage, openness to cooperation

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What is the aim measuring?

Do we measure anything? What? How often?

Use of money-based measurement

- Tracking changes (numbers)
- Value comparison (measurement unit)

The value is in the eye of the beholder. (strategy, synergy)

Quasi assets can not be sold on their own.

ScoreCards 1

1985 Skandia (Sweden)

- Five dimensions: finance, customers, employees, processes, renewal
- 73 traditional + 91 new (intellectual capital) ratios (164 ratios)

1992 Kaplan and Norton: Balanced Scorecard

- finance, customers, operational processes, learning-development

1993 Dow Chemical

- First in the USA

ScoreCards 2

Mid 1990s PWC

- ValueReporting: market, strategy, value creating activities, financial performance

1999 Edvinsson and Malone

- Skandia system corrected for redundancies
- 112 ratios remained even afterwards

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Thank you for your attention. Any questions?

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